Xidong WANG

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Ph.D. Candidate in Computer Science, The Chinese University of Hong Kong, Shenzhen
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Homepage / Google Scholar (**2.0K+ citations**) / GitHub

Research Interests

My goal is to democratize progress toward AGI through methods that improve model capabilities and computational efficiency together, enabling researchers with modest resources to participate in and extend this progress.

Architectures and algorithms for stronger models. Hybrid multimodal architectures (LongLLaVA), learning connections between Transformer blocks (T-Tuning), and reinforcement learning through value estimation and policy optimization (PPO+). T-Tuning and PPO+ are forthcoming work.

Earlier work: open, accessible, and scalable medical AI. Motivated by AI for social good: scalable alignment (ClinAlign), evaluation (CMB), multilingual models (Apollo, ApolloMoE), and open datasets and models in the Huatuo series, including Huatuo-26M.

Education

The Chinese University of Hong Kong, Shenzhen 2023 – Present
Ph.D. Candidate in Computer Science
Advisor: Prof. Benyou Wang

Beijing Institute of Technology 2019 – 2023
B.Eng. in Computer Science and Technology
Xuteli School – Elite talent program; Beijing, China

Research and Industry Experience

IQuest Research Mar. 2026 – Present
Research Intern: Foundation Model Post-training
Beijing, China

Contributing to STEM reasoning and agent capabilities, including IQuest-Q1. Blog / Model & code.

Ant Healthcare, Ant Group May 2025 – Feb. 2026
Research Intern: Rubric Reinforcement Learning
Hangzhou, China

Worked on rubric-based RL and the foundation model powering Ant Afu, an AI healthcare app in China with 150 million users (September 2026).

Shenzhen Research Institute of Big Data Aug. 2022 – Aug. 2023
Research Intern: Medical Language Models
Shenzhen, China

Contributed to the early development of the HuatuoGPT series.

Technical Reports

IQuest-Q1 Technical Report Blog

Publications

* *indicates equal contribution.* Impact figures reflect previously reported milestones.

1. **Efficiently Democratizing Medical LLMs for 50 Languages via a Mixture of Language Family Experts**
   Guorui Zheng*, **Xidong Wang***, Juhao Liang, Nuo Chen, Yuping Zheng, Benyou Wang
   ICLR – 2025 Paper

2. **LiveClin: A Live Clinical Benchmark without Leakage**
Xidong Wang, Shuqi Guo, Yue Shen, Junying Chen, Jian Wang, Jinjie Gu, Ping Zhang, Lei Liu, Benyou Wang
ICLR – 2026 Paper

3. **LongLLaVA: Scaling Multi-modal LLMs to 1000 Images Efficiently via a Hybrid Architecture**
Xidong Wang*, Dingjie Song*, Shunian Chen, Chen Zhang, Benyou Wang
Findings of EMNLP – 2025 Paper

4. **CMB: A Comprehensive Medical Benchmark in Chinese**
Xidong Wang, Guiming Hardy Chen, Dingjie Song, Zhiyi Zhang, Zhihong Chen, Qingying Xiao, Feng Jiang, Jianquan Li, Xiang Wan, Benyou Wang, Haizhou Li
NAACL – 2024 **Evaluation participation from 20+ research institutions and companies, including publicly listed firms.** Website Paper

5. **ClinAlign: Scaling Healthcare Alignment from Clinician Preference**
Shiwei Lyu*, **Xidong Wang***, Lei Liu, Hao Zhu, Chaohe Zhang, Jian Wang, Jinjie Gu, Benyou Wang, Yue Shen
Findings of ACL – 2026 Paper

6. **Huatuo-26M, a Large-scale Chinese Medical QA Dataset**
Xidong Wang*, Shunian Chen*, Jianquan Li*, Yuxuan Zhu, Xiangbo Wu, Zhiyi Zhang, Xiaolong Xu, Jie Fu, Prayag Tiwari, Xiang Wan, Benyou Wang
Findings of NAACL – 2025 Paper

7. **Beyond ViT Tokens: Masked-Diffusion Pretrained Convolutional Pathology Foundation Model for Cell-Level Dense Prediction**
Weiming Chen, Xitong Ling, Zhenyang Cai, **Xidong Wang**, Jiawen Li, Tian Guan, Benyou Wang, Yonghong He
NeurIPS – 2026 – Poster Paper

8. **Incentivizing Medical Vision Capabilities from Large-Scale Multimodal Pre-training**
Junying Chen, Zhenyang Cai, Yunjin Yang, Kunyuan Cai, Rongsheng Wang, **Xidong Wang**, Geyang Yu, Xiangyi Feng, Benyou Wang
NeurIPS – 2026 **Oral presentation** Website

9. **UniReason-Med: A Shared Grounded Reasoning Interface for 2D-to-3D Transfer in Medical VQA**
Mengzhuo Chen, Yan Shu, Chi Liu, Hongming Piao, **Xidong Wang**, Derek Li, Bryan Dai
EMNLP – 2026 – Main Conference Paper

10. **Hybrid Open-Ended Tri-Evolution Makes Better Deep Researcher**
Hongming Piao, Chi Liu, Mengzhuo Chen, Yan Shu, **Xidong Wang**, Derek Li, Ying Wei, Bryan Dai
EMNLP – 2026 – Main Conference Paper

11. **Apollo: A Lightweight Multilingual Medical LLM towards Democratizing Medical AI to 6B People**
Xidong Wang*, Nuo Chen*, Junyin Chen*, Yan Hu, Yidong Wang, Xiangbo Wu, Anningzhe Gao, Xiang Wan, Haizhou Li, Benyou Wang
Preprint – 2024 **120K+ downloads on Hugging Face** Paper

12. **HuatuoGPT-II, One-stage Training for Medical Adaptation of LLMs**
Junying Chen, **Xidong Wang**, Anningzhe Gao, Feng Jiang, Shunian Chen, Hongbo Zhang, Dingjie Song, Wenya Xie, Chuyi Kong, Jianquan Li, Xiang Wan, Haizhou Li, Benyou Wang
COLM – 2024 Paper

13. **HuatuoGPT-o1, Towards Medical Complex Reasoning with LLMs**
Junying Chen, Zhenyang Cai, Ke Ji, **Xidong Wang**, Wanlong Liu, Rongsheng Wang, Jianye Hou, Benyou Wang
Findings of ACL – 2025 Paper

14. **HuatuoGPT-Vision, towards Injecting Medical Visual Knowledge into Multimodal LLMs at Scale**
Junying Chen, Ruyi Ouyang, Anningzhe Gao, Shunian Chen, Guiming Hardy Chen, **Xidong Wang**, Ruifei Zhang, Zhenyang Cai, Ke Ji, Guangjun Yu, Xiang Wan, Benyou Wang
EMNLP – 2024 **824K+ downloads on Hugging Face** Paper

15. **CoD, towards an Interpretable Medical Agent using Chain of Diagnosis**
Junying Chen, Chi Gui, Anningzhe Gao, Ke Ji, **Xidong Wang**, Xiang Wan, Benyou Wang
Findings of ACL – 2025 Paper

16. **ShareGPT-4o-Image: Aligning Multimodal Models with GPT-4o-Level Image Generation**
Junying Chen, Zhenyang Cai, Pengcheng Chen, Shunian Chen, Ke Ji, **Xidong Wang**, Yunjin Yang, Benyou Wang
Preprint – 2025 Paper

17. **XtraGPT: LLMs for Human-AI Collaboration on Controllable Academic Paper Revision**
Nuo Chen, Andre Lin HuiKai, Jiaying Wu, Junyi Hou, Zining Zhang, Qian Wang, **Xidong Wang**, Bingsheng He
ACL – 2026 Paper

18. **Roadmap towards Superhuman Speech Understanding using Large Language Models**
Fan Bu, Yuhao Zhang, **Xidong Wang**, Benyou Wang, Qun Liu, Haizhou Li
Preprint – 2024 Paper

19. **Do Phone-Use Agents Respect Your Privacy?**
Zhengyang Tang*, Ke Ji*, **Xidong Wang***, Zihan Ye*, Xinyuan Wang, Yiduo Guo, Ziniu Li, Chenxin Li, Jingyuan Hu, Shunian Chen, Tongxu Luo, Jiaxi Bi, Zeyu Qin, Shaobo Wang, Xin Lai, Pengyuan Lyu, Junyi Li, Can Xu, Chengquan Zhang, Han Hu, Ming Yan, Benyou Wang
Preprint – 2026 Paper

20. **WaveMind: Towards a Conversational EEG Foundation Model Aligned to Textual and Visual Modalities**
Ziyi Zeng, Zhenyang Cai, Yixi Cai, **Xidong Wang**, Junying Chen, Rongsheng Wang, Yipeng Liu, Siqi Cai, Benyou Wang, Zhiguo Zhang, Haizhou Li
Preprint – 2025 Paper

21. **To What Extent Do Token-Level Representations from Pathology Foundation Models Improve Dense Prediction?**
Weiming Chen, Xitong Ling, **Xidong Wang**, Zhenyang Cai, Yijia Guo, Mingxi Fu, Ziyi Zeng, Minxi Ouyang, Jiawen Li, Yizhi Wang, Tian Guan, Benyou Wang, Yonghong He
Preprint – 2026 Paper

22. **Less is More: A Simple yet Effective Token Reduction Method for Efficient Multi-modal LLMs**
Dingjie Song, Wenjun Wang, Shunian Chen, **Xidong Wang**, Michael X. Guan, Benyou Wang
COLING – 2025 Paper

23. **MLLM-Bench: Evaluating Multi-modal LLMs using GPT-4V**
Wentao Ge, Shunian Chen, Guiming Hardy Chen, Nuo Chen, Junying Chen, Zhihong Chen, Wenya Xie, Shuo Yan, Chenghao Zhu, Ziyue Lin, Dingjie Song, **Xidong Wang**, Anningzhe Gao, Zhiyi Zhang, Jianquan Li, Xiang Wan, Benyou Wang
NAACL – 2025 Paper

24. **LLMs for Doctors: Leveraging Medical LLMs to Assist Doctors, Not Replace Them**
Wenya Xie, Qingying Xiao, Yu Zheng, **Xidong Wang**, Junying Chen, Ke Ji, Anningzhe Gao, Prayag Tiwari, Xiang Wan, Feng Jiang, Benyou Wang
npj Health Systems – 2025 Paper

Teaching

CSC6052: Natural Language Processing Spring 2024 & 2025
Teaching Assistant, CUHK-Shenzhen. Instructor: Prof. Benyou Wang.

CSC6203: Large Language Models Fall 2023
Teaching Assistant, CUHK-Shenzhen. Instructor: Prof. Benyou Wang.

Service, Mentoring and Talks

Reviewer: ICLR (2025–2027), ICML (2026), NeurIPS (2026), COLM (2025–2026), TMLR, and ACL Rolling Review.
Open source: Lead maintainer of Awesome-AI4Med (**2.9K+ GitHub stars**).
Mentoring: Guorui Zheng (2024), MSE student at CUHK-Shenzhen; subsequently an Algorithm Engineer at ByteDance.
Invited talk: The Fin AI, December 2024. "Efficiently Democratizing Medical LLMs for 50 Languages via a Mixture of Language Family Experts."